DATA SCIENCE INTERNSHIP

WEEK 1 ASSIGNMENT REPORT

E-COMMERCE DATA CLEANING AND SALES DATA ANALYSIS

Prepared for:

WeIntern Pvt Ltd

Submitted by:

Ishita Misra

Internship Domain:

Data Science

Assignment for Week 1 –

Data Cleaning, Sales Analysis and Data Visualization

Tools & Technologies:

Python | Pandas | NumPy | Matplotlib | Seaborn | Jupyter Notebook

Submission Date:

14 October 2026

TABLE OF CONTENTS

S. No.	Contents	Page No.
1	Title and Objective	1
2	Dataset Overview	2–3
3	Cleaning Summary	4
4	KPI Summary	5
5	Trend Analysis	6
6	Customer Insights	7
7	Product and Category Analysis	8
8	Payment and Delivery Analysis	9
9	Regional and Geographic Analysis	10
10	Data Visualisation	11
11	Key Findings and Business Insights	12–13
12	Limitations and Recommendations	14

1. Objective

The objective of this project is to analyze an e-commerce sales dataset to understand sales performance, customer purchasing behavior, product and category performance, and monthly sales trends. The project focuses on examining the dataset, identifying and addressing data quality issues, calculating key performance indicators (KPIs), and presenting meaningful insights through statistical summaries and visualizations.

The analysis aims to identify the top-performing product categories and products, understand repeat customer purchasing patterns, and highlight changes in sales over time. The findings will be used to develop practical, data-driven recommendations to support business decision-making and improve sales performance.

The project follows a structured data analysis workflow involving data inspection, cleaning, validation, exploratory data analysis, KPI calculation, visualization, and interpretation of results. The final deliverables include a cleaned dataset, a Jupyter Notebook containing the analysis, a data quality audit, a KPI summary workbook, and a final report documenting the key findings and recommendations.

2. Dataset Overview

The cleaned e-commerce dataset was examined to determine its size, field types, and date coverage. It contains 10,009 rows and 16 columns, covering order details, customers, products, sales amounts, payment methods, delivery status, and geographic information.

Dataset Summary

Attribute	Details
Dataset file	ecommerce_cleaned_dataset.xlsx
Worksheet	Cleaned_Data
Total rows	10,009
Total columns	16
Date field	order_date
Earliest valid order date	3 January 2014
Latest valid order date	15 January 2035
Valid date values	9,942
Missing date values	67

Column Names and Data Types

Column	Data type	Description
order_id	Object (text)	Order identifier
order_date	Datetime	Date associated with the order
customer_id	Object (text)	Customer identifier
customer_name	Object (text)	Customer name
product_id	Object (text)	Product identifier
product_name	Object (text)	Product name
category	Object (text)	Main product category
subcategory	Object (text)	Product subcategory
quantity	Float	Quantity recorded for the order line
unit_price	Float	Price per unit
total_amount	Float	Recorded total amount
discount	Float	Discount value
payment_mode	Object (text)	Payment method
delivery_status	Object (text)	Delivery status
city	Object (text)	Customer or delivery city
region	Object (text)	Geographic region

Overview and Initial Observations

The dataset contains a combination of categorical fields, numerical fields, and a date field, making it suitable for sales performance analysis, product and category comparisons, customer purchasing analysis, and time-based trend analysis.

The order_id, customer_id, and product_id fields support the identification of orders, customers, and products. Numerical fields such as quantity, unit_price, total_amount, and discount can be used to calculate sales-related metrics. The order_date field enables monthly trend analysis, while category, subcategory, payment_mode, city, and region allow comparisons across different business segments.

Important data quality observation: Although the dataset contains 10,009 rows, only 9,942 rows have a valid order_date, leaving 67 missing date values. The observed date range also extends from 2014 to 2035. These dates should be validated before interpreting monthly trends, since unusually distant dates may affect the analysis.

Note: The data types above are those detected when the cleaned Excel file was loaded into pandas. The date range reflects valid dates only; missing dates are excluded.

3. Cleaning Summary

The e-commerce dataset was cleaned and prepared for analysis to improve data consistency, reliability, and suitability for calculating key performance indicators (KPIs). The cleaning process focused on examining missing values, checking duplicate records, validating data types, and reviewing fields required for sales and customer analysis.

The cleaned dataset contains 10,009 rows and 16 columns. The order_date field contains 67 missing values, with 9,942 valid dates ranging from 3 January 2014 to 15 January 2035. These missing dates limit the number of records that can be used for time-based analysis and should be considered when interpreting monthly sales trends.

The dataset includes numerical fields such as quantity, unit_price, total_amount, and discount, categorical fields such as category, subcategory, payment_mode, and delivery_status, and identifiers such as order_id, customer_id, and product_id. These fields were retained to support sales calculations, product comparisons, and customer-level analysis.

The cleaning and validation process should also account for duplicate records, missing or inconsistent identifiers, invalid numerical values, and dates outside the expected business period. Records should only be removed or modified when the relevant validation rules justify the decision. Missing dates should not be replaced with invented values, and unusual dates should be flagged for review rather than automatically discarded.

The cleaned dataset serves as the input for the subsequent KPI calculations, monthly trend analysis, customer insights, and visualizations. Any remaining data quality issues and assumptions should be documented in the data quality audit so that the results can be interpreted transparently.

4. KPI Summary

The key performance indicators (KPIs) provide a quantitative overview of e-commerce sales performance, order activity, product demand, and customer purchasing behavior. The primary metrics for this analysis are total amount, unique orders, average order value, units sold, top-performing category, and top-performing product.

Total Amount: This metric represents the sum of values recorded in the total_amount column. It provides an overview of the monetary value represented in the dataset. Since the dataset does not independently establish whether these amounts reflect completed transactions, discounts, refunds, or realized revenue, the metric is referred to as total amount rather than confirmed revenue.

Unique Orders: The number of distinct order identifiers in the order_id column measures the number of uniquely identified orders. This metric helps assess order activity and provides the denominator for calculating average order value.

Average Order Value (AOV): Average order value is calculated by dividing the total amount by the number of unique orders. It indicates the average recorded amount per order. This calculation assumes that total_amount is comparable across records and that each distinct order should be counted once.

Units Sold: The sum of the quantity column represents the total recorded quantity across the dataset. This metric helps measure product demand, provided the quantity values are valid and consistently defined.

Top-Performing Category: Product categories are compared by aggregating total_amount for each category. The category with the highest aggregate amount is identified as the leading category by recorded monetary value.

Top-Performing Product: Products are compared using their aggregate total_amount. The product with the highest total is identified as the leading product by recorded monetary value.

These KPIs establish a baseline for evaluating sales performance and support further investigation into monthly trends, customer purchasing behavior, and differences across product categories. The numerical KPI results should be taken directly from the calculations in the analysis notebook and interpreted alongside the data quality audit to ensure that the findings accurately reflect the cleaned dataset.

5. Trend Analysis

The monthly sales trend analysis examines changes in recorded sales amounts over time using the order_date and total_amount fields. The objective is to identify periods of stronger or weaker sales activity, recurring patterns, and unusual fluctuations that may warrant further investigation.

The cleaned dataset contains 9,942 valid order dates, ranging from 3 January 2014 to 15 January 2035. Records with missing dates cannot be assigned reliably to a calendar month and should therefore be excluded from monthly trend calculations unless their dates can be verified from another reliable source.

For this analysis, the recorded total amount is aggregated by month to examine changes in monetary value over time. Monthly order activity can also be assessed using distinct order identifiers, while the quantity field can be aggregated to understand changes in recorded product demand.

The monthly trend analysis should focus on identifying the highest and lowest months by recorded amount, periods of growth or decline, and significant month-to-month fluctuations. Any apparent seasonal patterns should be interpreted cautiously, as the unusually broad date range may include inconsistent or unexpected dates that could distort the results.

The findings from this analysis can help identify periods that deserve closer examination and support more informed decisions about sales planning, inventory management, and promotional activities. All conclusions should be based on the monthly calculations and visualizations generated from the cleaned dataset rather than assumptions about expected sales behavior.

6. Customer Insights

Customer purchasing behaviour was analysed using the customer_id, customer_name, order_id, quantity, and total_amount columns. These fields support an assessment of customer activity, purchase frequency, and spending patterns.

The analysis focuses on the following areas:

- Unique customers: Count of distinct customer IDs to estimate the number of customers represented in the dataset.
- Repeat customers: Identification of customers associated with more than one distinct order.
- Customer spending: Aggregation of total_amount by customer to identify customers contributing the highest recorded transaction amounts.
- Average spending per customer: Calculation of total recorded transaction amount divided by the number of unique customers.
- Purchase frequency: Comparison of the number of distinct orders associated with each customer.
- Customer contribution: Examination of how recorded transaction amounts are distributed across customers.

The customer_id field is used as the primary identifier for customer-level analysis, while customer_name provides a readable label. Using customer IDs helps reduce the risk of treating customers with similar names as the same person.

The dataset contains 10,009 rows, but this represents transaction records rather than a confirmed count of unique customers. Actual customer counts, repeat-purchase rates, spending averages, and the highest-spending customers should be reported using the calculated results from the analysis notebook. These figures cannot be established from the dataset's row count alone.

7. Product and Category Analysis

Product and category performance was examined using the product_id, product_name, category, subcategory, quantity, and total_amount columns. These fields support comparisons of recorded transaction amounts and quantities across products and product categories.

The analysis focuses on the following areas:

- Category performance: Aggregation of total_amount by category to identify categories with the highest recorded transaction amounts.
- Subcategory performance: Comparison of subcategories to understand their contribution to overall transaction amounts.
- Top-performing products: Ranking products by aggregated total_amount to identify products associated with the highest recorded amounts.
- Quantity sold: Summation of quantity by product and category to compare sales volume.
- Product contribution: Assessment of each product's share of the total recorded transaction amount.
- Revenue distribution: Comparison of transaction amounts across categories to identify how recorded sales value is distributed.

The product_id field provides a product identifier, while product_name makes the results easier to interpret. Category and subcategory groupings help reveal differences in performance across product segments.

The analysis should distinguish between sales value and sales volume. A product with a high total transaction amount may not have the highest quantity sold, while a product sold in large quantities may contribute less to the overall amount.

The dataset includes the required product, category, quantity, and transaction-amount fields for this analysis. However, the actual highest-performing products and categories, their totals, and their percentage contributions must be taken from the calculated notebook results; these rankings cannot be confirmed from the available summary statistics alone.

8. Payment and Delivery Analysis

Payment and delivery patterns were examined using the payment_mode, delivery_status, order_id, and total_amount columns. These fields support an assessment of customer payment preferences and the distribution of orders across delivery statuses.

The analysis focuses on the following areas:

- Payment mode distribution: Counting orders by payment mode to identify the most frequently recorded payment methods.
- Payment value comparison: Aggregating total_amount by payment mode to compare the transaction amounts associated with different methods.
- Delivery status distribution: Counting orders by delivery status to understand how orders are represented across the recorded statuses.
- Transaction amount by delivery status: Comparing aggregated transaction amounts across delivery statuses.
- Order status proportions: Calculating the percentage of orders associated with each recorded delivery status.

The order_id field is used to distinguish orders, while payment_mode and delivery_status provide the grouping variables for the analysis.

9. Regional and Geographic Analysis

Geographic sales patterns were examined using the city, region, order_id, and total_amount columns. These fields support comparisons of recorded transaction amounts and order distribution across different geographic locations.

The analysis focuses on the following areas:

- Regional performance: Aggregation of total_amount by region to identify regions with the highest recorded transaction amounts.
- City-level performance: Comparison of cities based on their aggregated transaction amounts.
- Order distribution: Counting distinct orders across regions and cities to understand the geographic distribution of orders.
- Quantity distribution: Comparing total quantities across regions to examine differences in recorded sales volume.
- Regional contribution: Calculating each region's percentage share of the total recorded transaction amount.
- Geographic concentration: Identifying locations that contribute relatively larger or smaller amounts to the overall dataset.

The region column provides a broader geographic grouping, while city supports more detailed comparisons. Analysing both fields helps distinguish regional patterns from city-specific performance.

The analysis can help identify locations that may warrant further business attention. However, higher recorded transaction amounts do not necessarily indicate higher profitability because the dataset does not establish costs, operating expenses, or profit margins.

Actual top-performing regions, cities, order counts, and percentage contributions should be reported from the notebook's calculated results.

10. Data Visualisation

Data visualisation was used to present the analysis results in a clear and interpretable format. Charts help identify patterns, compare categories, and communicate findings that may not be immediately apparent from tabular data.

The visualisation section covers the following areas:

- Monthly sales trend: A line chart showing aggregated transaction amounts over time to identify fluctuations and longer-term patterns.
- Category performance: A bar chart comparing total transaction amounts across product categories.
- Top-performing products: A bar chart displaying products with the highest aggregated transaction amounts.
- Customer analysis: A chart comparing customer spending or purchase frequency to highlight differences in customer activity.
- Payment mode distribution: A bar chart showing the number of orders recorded for each payment mode.
- Delivery status distribution: A chart comparing order counts across the recorded delivery statuses.
- Regional performance: A bar chart comparing transaction amounts across regions or cities.

The charts use the relevant dataset columns and aggregation methods to support the corresponding analysis. Monthly trend visualisations exclude records with missing order_date values because those records cannot be assigned reliably to a month.

Chart labels, titles, and axes should clearly identify the measure being presented. For example, a chart showing total_amount should be labelled as a transaction-amount comparison rather than confirmed profit.

The visualisations complement the numerical summaries by making trends, differences, and distributions easier to interpret. Actual chart findings should be described using the patterns visible in the generated plots rather than assumed in advance.

11. Key Findings and Business Insights

The e-commerce dataset provides a foundation for examining transaction patterns, customer behaviour, product performance, payment preferences, delivery status, and geographic distribution.

The analysis supports the following areas of business insight:

- Sales performance: Monthly aggregation of total_amount can help identify periods with relatively high or low recorded transaction amounts.
- Product strategy: Category and product comparisons can highlight which products and categories contribute the most to recorded transaction value and quantity.
- Customer retention: Repeat-purchase analysis can help distinguish customers who place multiple orders from those associated with only one order.
- Payment preferences: Payment mode comparisons can reveal which payment methods are most frequently recorded.
- Delivery monitoring: Delivery status counts can help identify the distribution of orders across the available status categories.
- Geographic opportunities: Regional and city-level comparisons can identify locations with relatively high or low recorded transaction amounts.
- Data quality awareness: Missing order dates and the unusually broad date range should be considered when interpreting trends and drawing conclusions.

These insights can inform further investigation into sales patterns, customer engagement, product selection, and order fulfilment. However, business recommendations should be based on the actual numerical results and visualisations produced by the analysis notebook.

Since the dataset's numerical findings have not been established in the available summary, no specific product, customer, payment method, region, or month can yet be identified as the strongest performer. The final findings should name the relevant categories and include their calculated values wherever available.

12. Limitations and Recommendations

Limitations

The analysis has several limitations that should be considered when interpreting the results:

- Missing dates: The dataset contains 67 records with missing order_date values. These records cannot be included reliably in date-based trend analysis without additional information.
- Unusual date range: The available dates range from 3 January 2014 to 15 January 2035. This unusually broad range should be validated before drawing conclusions about historical or future sales trends.
- Data completeness: Missing or inconsistent values in relevant fields may affect customer, product, payment, delivery, and geographic comparisons.
- Transaction interpretation: The total_amount column represents recorded transaction amounts, but the dataset alone does not establish whether these amounts reflect completed sales, cancellations, refunds, or net revenue.
- Profitability: The available fields do not provide sufficient information to calculate profit margins or net profit because comprehensive cost and expense data are not available.
- Customer identification: Customer-level analysis depends on the consistency and completeness of customer_id. Missing or inconsistent identifiers may affect customer counts and repeat-purchase calculations.
- Descriptive analysis: The findings describe patterns in the available dataset. They do not establish causal relationships or guarantee future business performance.

Recommendations

Based on these limitations, the following actions are recommended:

1. Validate date values: Investigate missing dates and confirm whether unusually late dates represent legitimate records or data-entry errors.
2. Maintain data quality checks: Regularly check for missing values, duplicate records, inconsistent categories, invalid numerical values, and identifier inconsistencies.
3. Monitor sales trends: Use monthly transaction summaries to track changes over time after validating the date range.

4. Evaluate product performance: Compare both transaction amounts and quantities when identifying products and categories for further investigation.
5. Study repeat customers: Use customer-level order counts and spending summaries to investigate purchasing frequency and customer retention.
6. Review delivery statuses: Examine the distribution of delivery statuses and investigate records that may require operational follow-up.
7. Improve financial reporting: Confirm the meaning of total_amount and incorporate reliable cost, refund, and cancellation information before making profitability claims.
8. Automate the workflow: Reuse the data-cleaning and analysis notebook to make future updates more consistent and reproducible.

These recommendations aim to improve data reliability and support more informed business decisions. Their implementation should be guided by the findings established through the completed analysis.